

THEORY



Training Program 01

Advanced NLP & Large Language Models (LLMs)



CONFIANZA23
Inteligencia y Seguridad

Training Program 01

M01: Text Preprocessing

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects



MODULE M01

TEXT PREPROCESSING

CHAPTER M10

RAG & Real-World Projects



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10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

10.2 RAG Architectures: Retrieval-Augmented Generation Systems

10.3 Real-World Applications of LLMs and RAG

10.4 Presentation Guidelines and Final Review

10.5 Student Capstone Project Presentations

10.6 Course Wrap-up and Next Steps

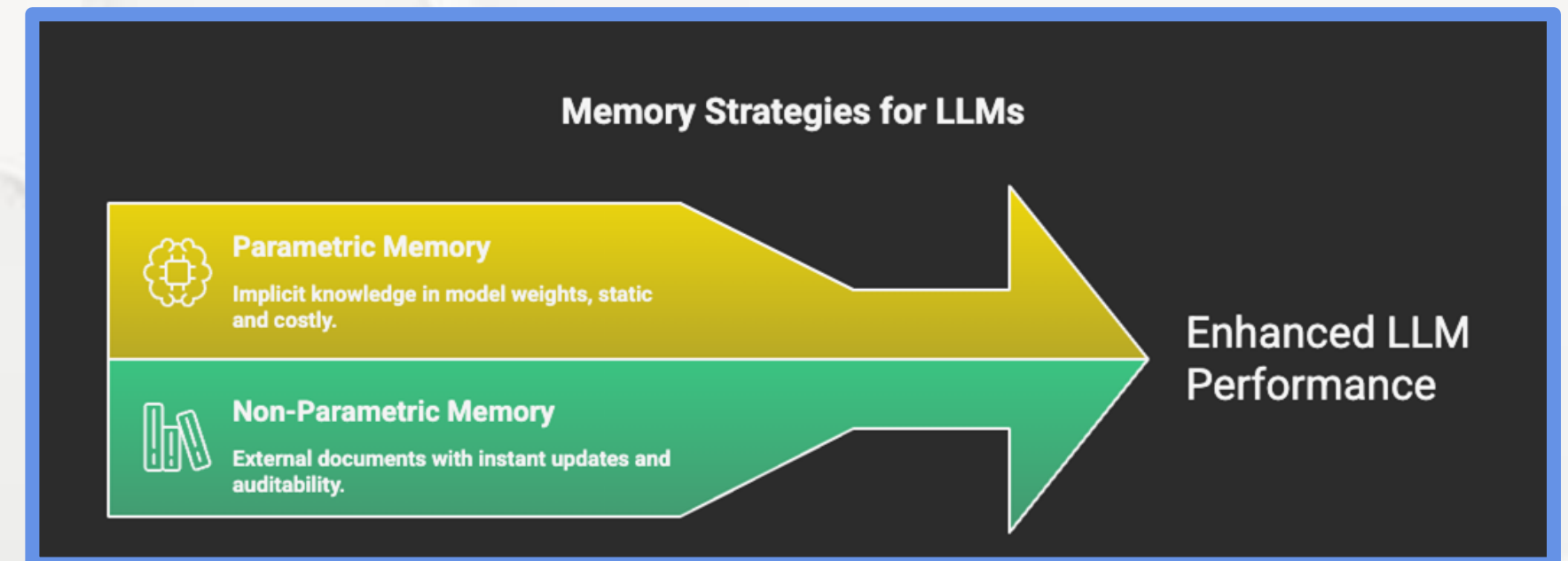


Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

Why RAG Exists

Large Language Models are powerful reasoning engines, but their knowledge is frozen in time. Retrieval-Augmented Generation (RAG) solves this limitation by connecting models to real, up-to-date, and private information sources.

- LLMs suffer from a knowledge cut-off.
- They cannot access private or proprietary data.
- Fine-tuning is not designed to inject new facts.
- RAG enables real-time knowledge access without retraining.

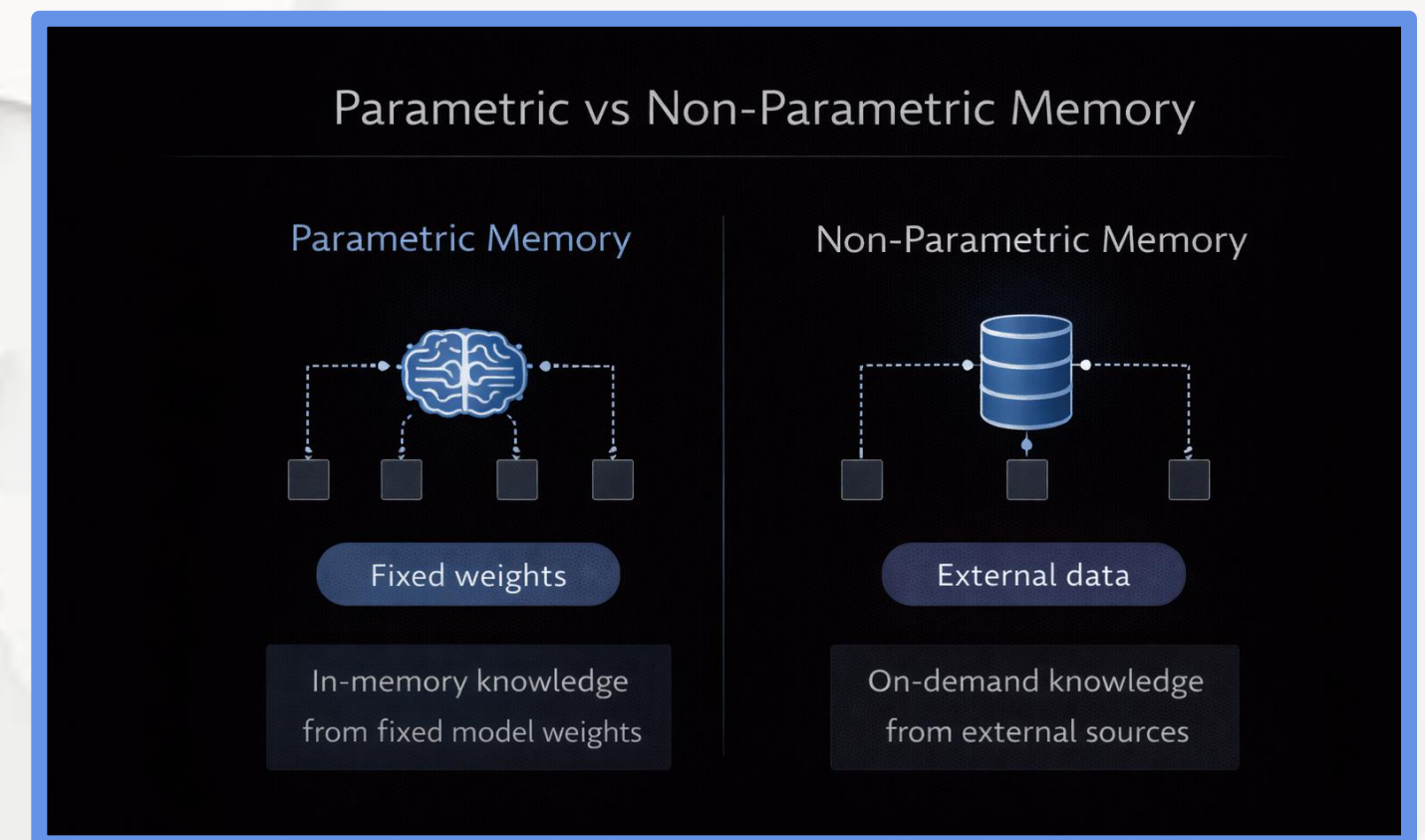


Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

Parametric vs Non-Parametric Memory

LLMs store knowledge implicitly in weights, while RAG introduces explicit, searchable memory. This distinction is critical for reliability, updates, and compliance.

- Parametric memory: knowledge encoded in model weights.
- Static and expensive to update.
- Non-parametric memory: external documents + retrieval.
- Instantly updateable and auditable.



Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

What Retrieval-Augmented Generation Is

RAG transforms the model from a “knower” into a “reader”, forcing answers to be grounded in retrieved documents visible in the context window.

- Model answers using retrieved documents.
- Reduces hallucinations.
- Enables citations and traceability.
- Improves trust and explainability.

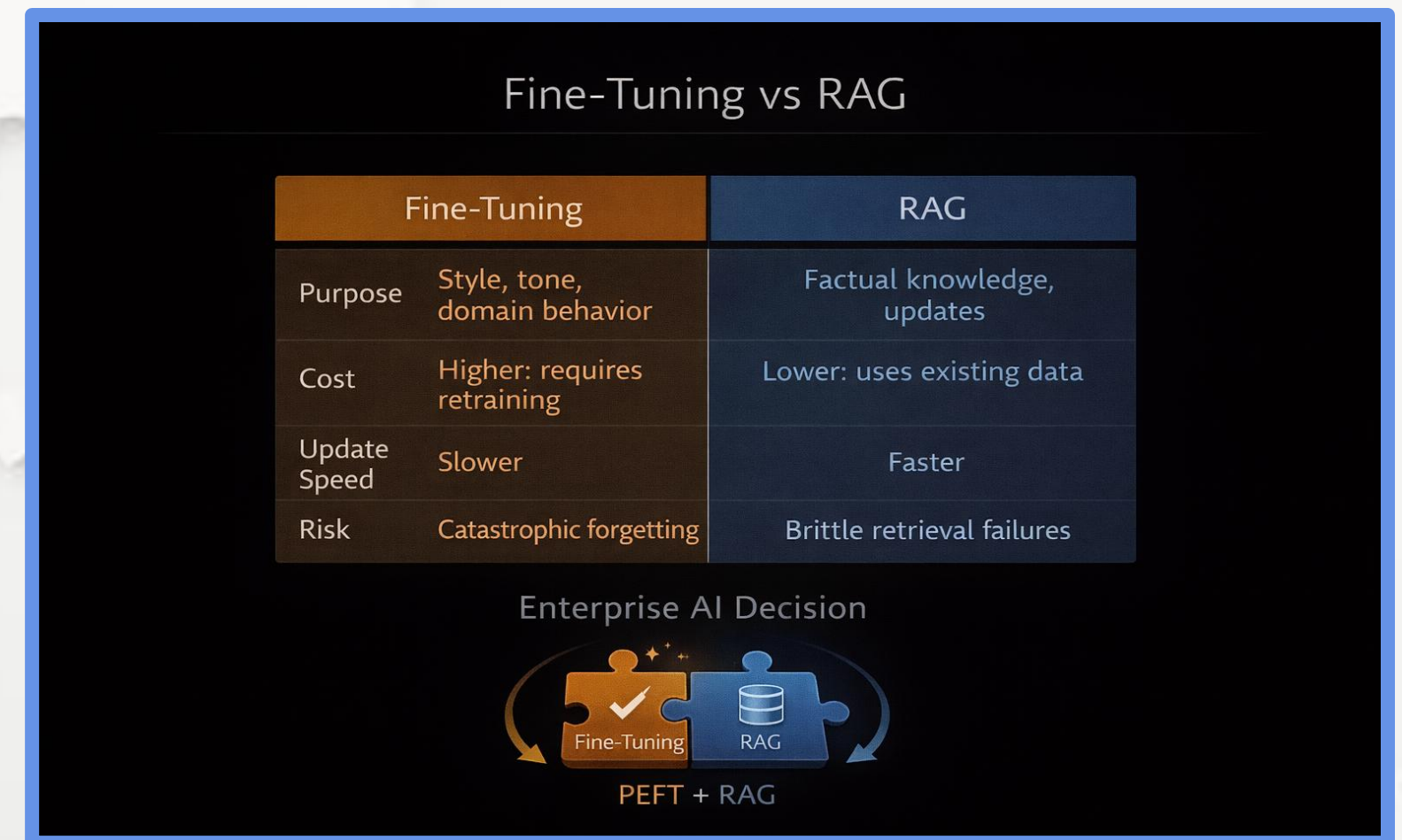


Chapter 10.1 Review and Introduction to Retrieval-Augmented Generation (RAG)

RAG vs Fine-Tuning

RAG and fine-tuning solve different problems. Confusing them leads to brittle systems. The strongest architectures use both strategically.

- Fine-tuning: style, tone, domain behavior.
- RAG: factual knowledge and updates.
- RAG avoids catastrophic forgetting.
- Enterprise systems often combine PEFT + RAG.



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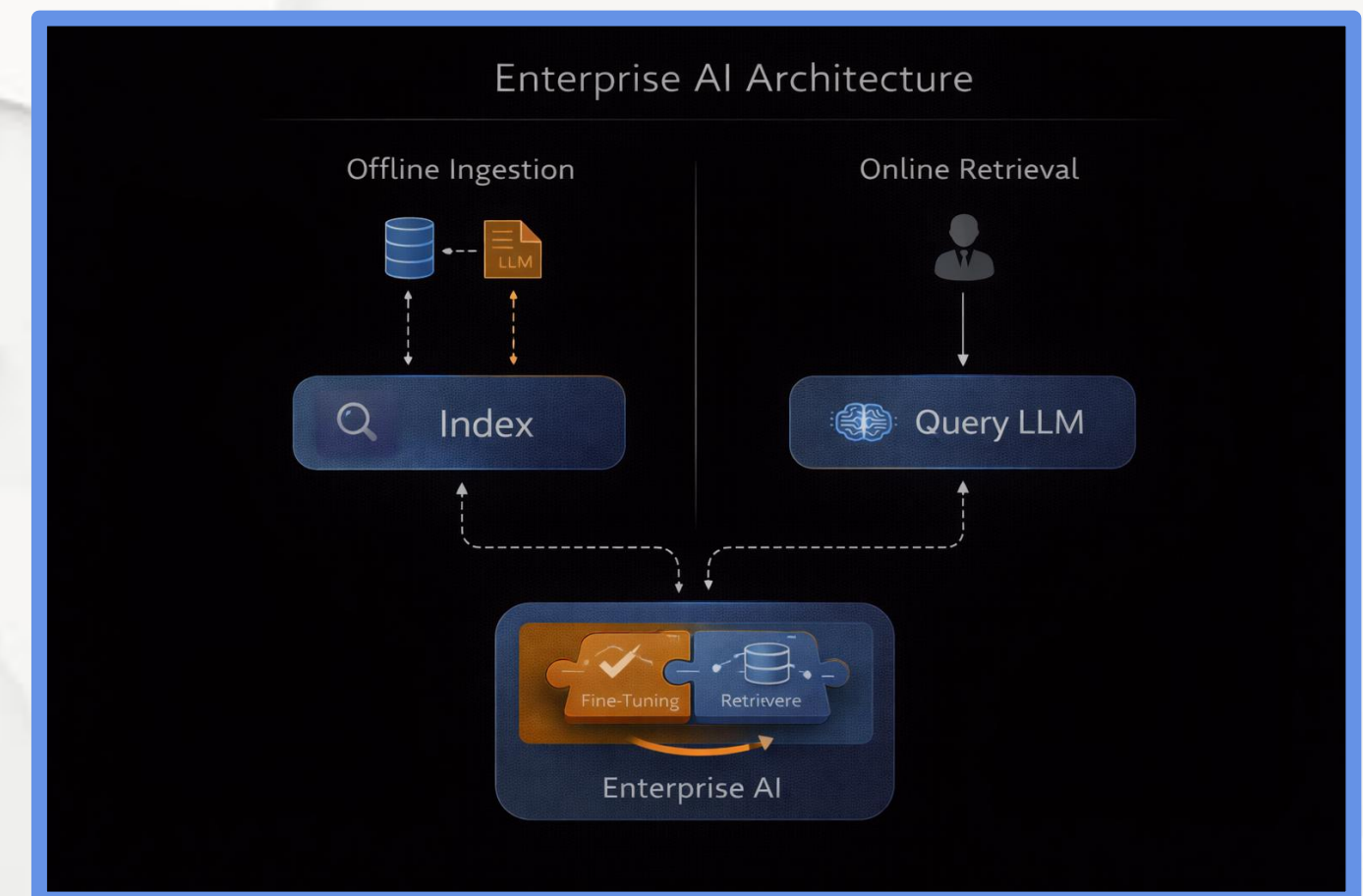


Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

The Hallucination Problem

LLMs are optimized for fluency, not truth. Hallucinations occur when the model fills gaps probabilistically instead of relying on verified information.

- Fluent $\neq$ factual.
- Hallucinations scale with confidence.
- Dangerous in legal, medical, and financial domains.
- Retrieval acts as a grounding constraint.

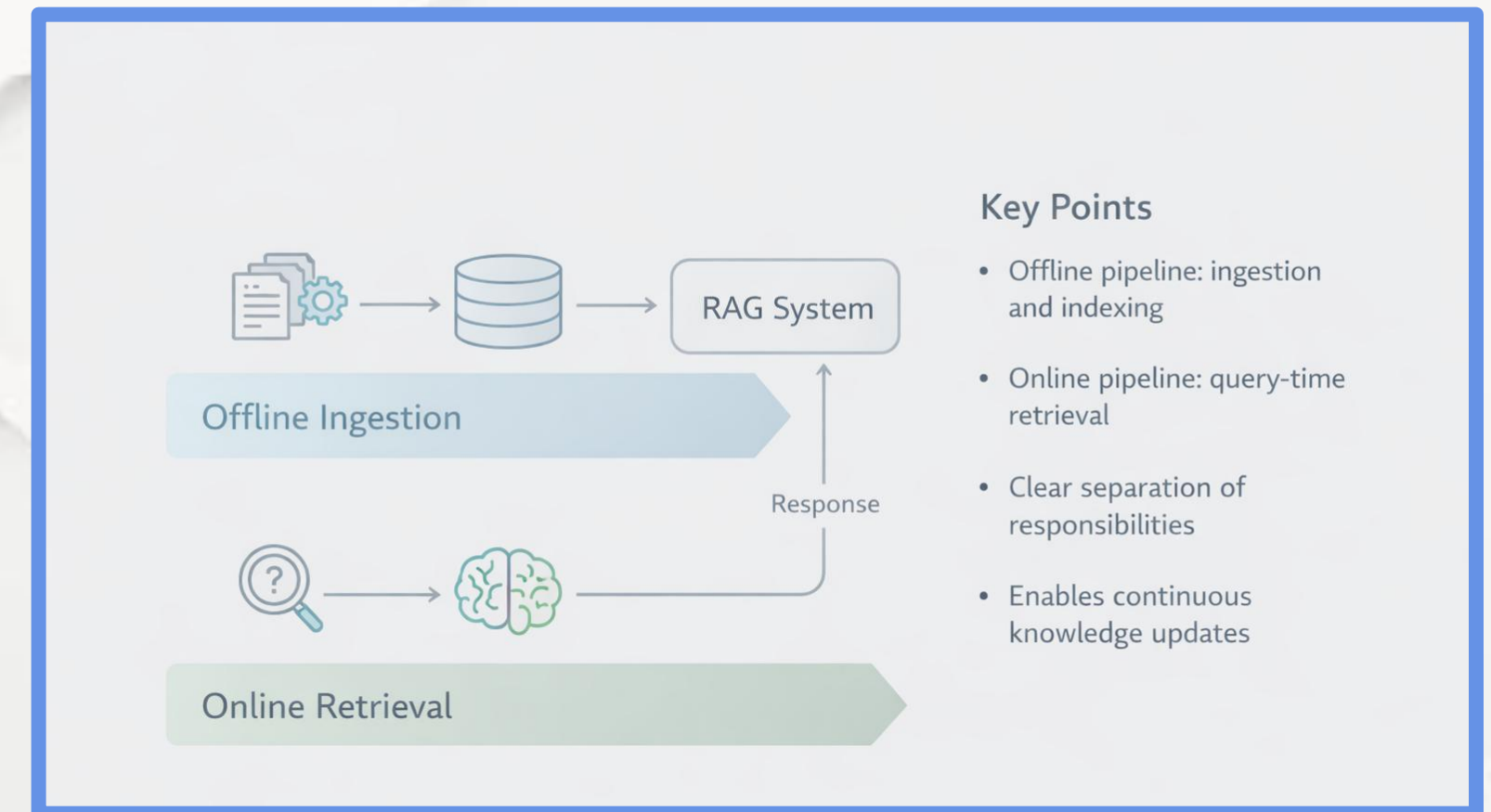


Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

The Dual-Pipeline RAG Architecture

A production RAG system separates knowledge preparation from real-time reasoning to ensure scalability and performance.

- Offline pipeline: ingestion and indexing.
- Online pipeline: query-time retrieval.
- Clear separation of responsibilities.
- Enables continuous knowledge updates.

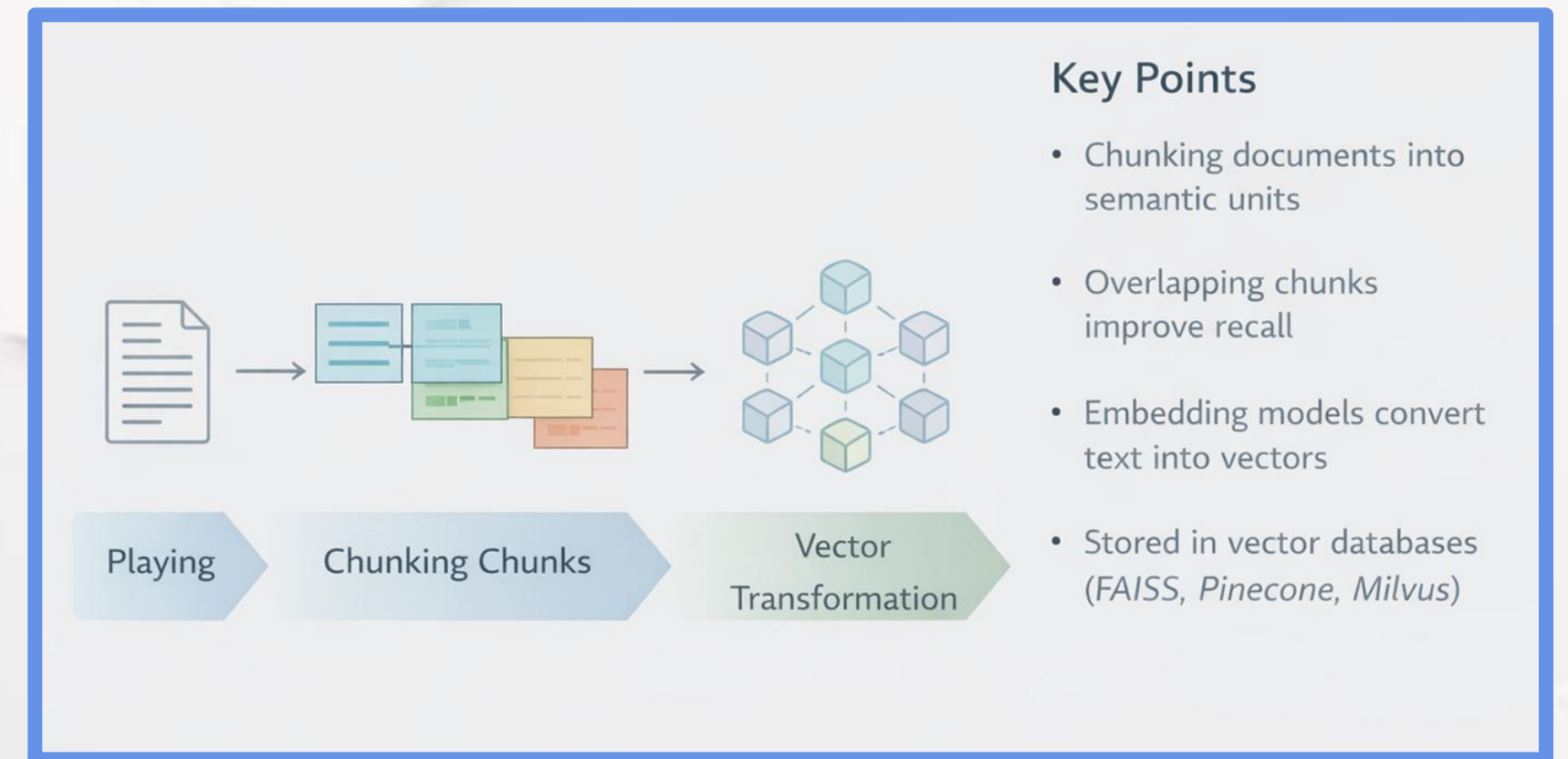


Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

Indexing Pipeline: Chunking & Embeddings

Before retrieval is possible, documents must be transformed into searchable vector representations.

- Chunking documents into semantic units.
- Overlapping chunks improve recall.
- Embedding models convert text into vectors.
- Stored in vector databases (FAISS, Pinecone, Milvus).

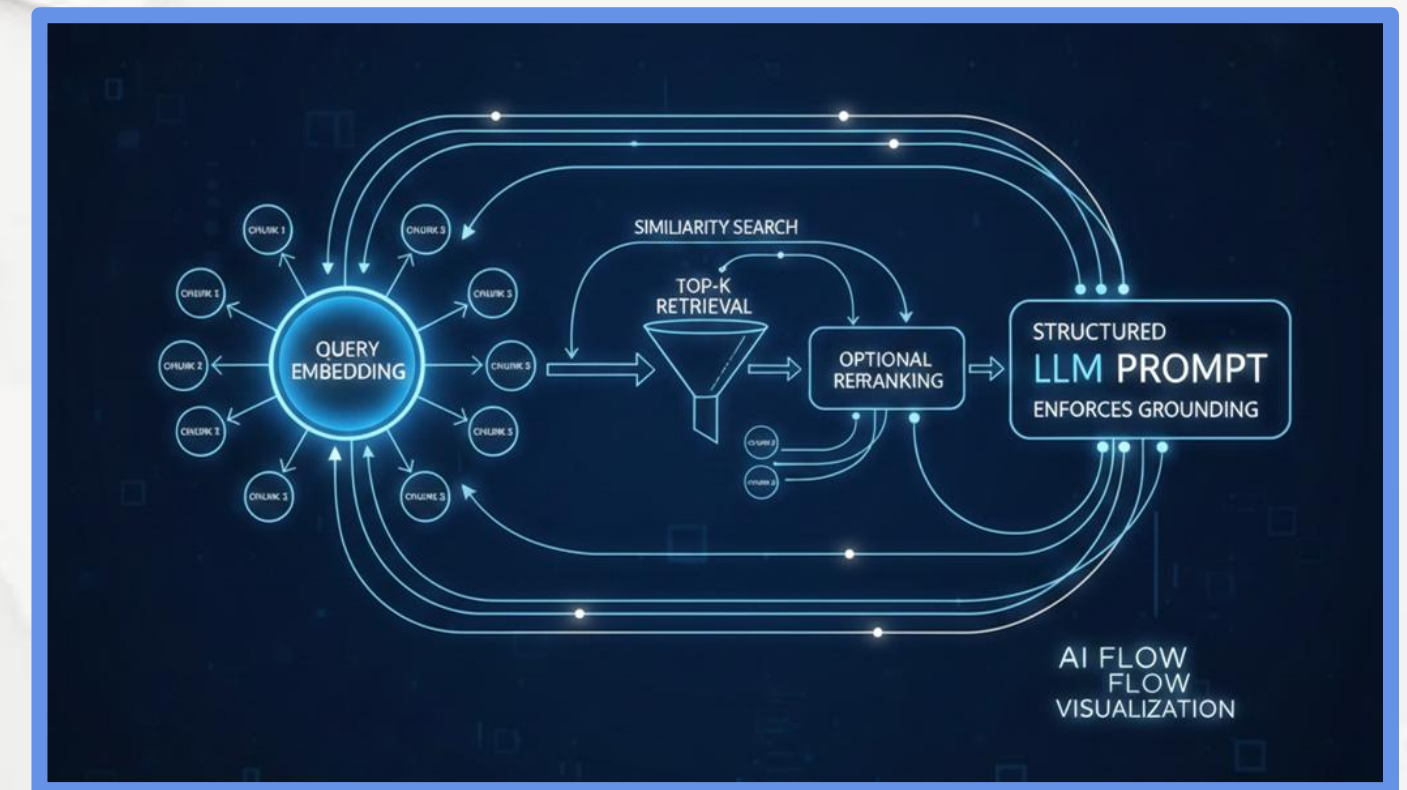


Chapter 10.2 RAG Architectures: Retrieval-Augmented Generation Systems

Retrieval and Prompt Augmentation

At query time, the system retrieves the most relevant chunks and injects them into the prompt to guide generation.

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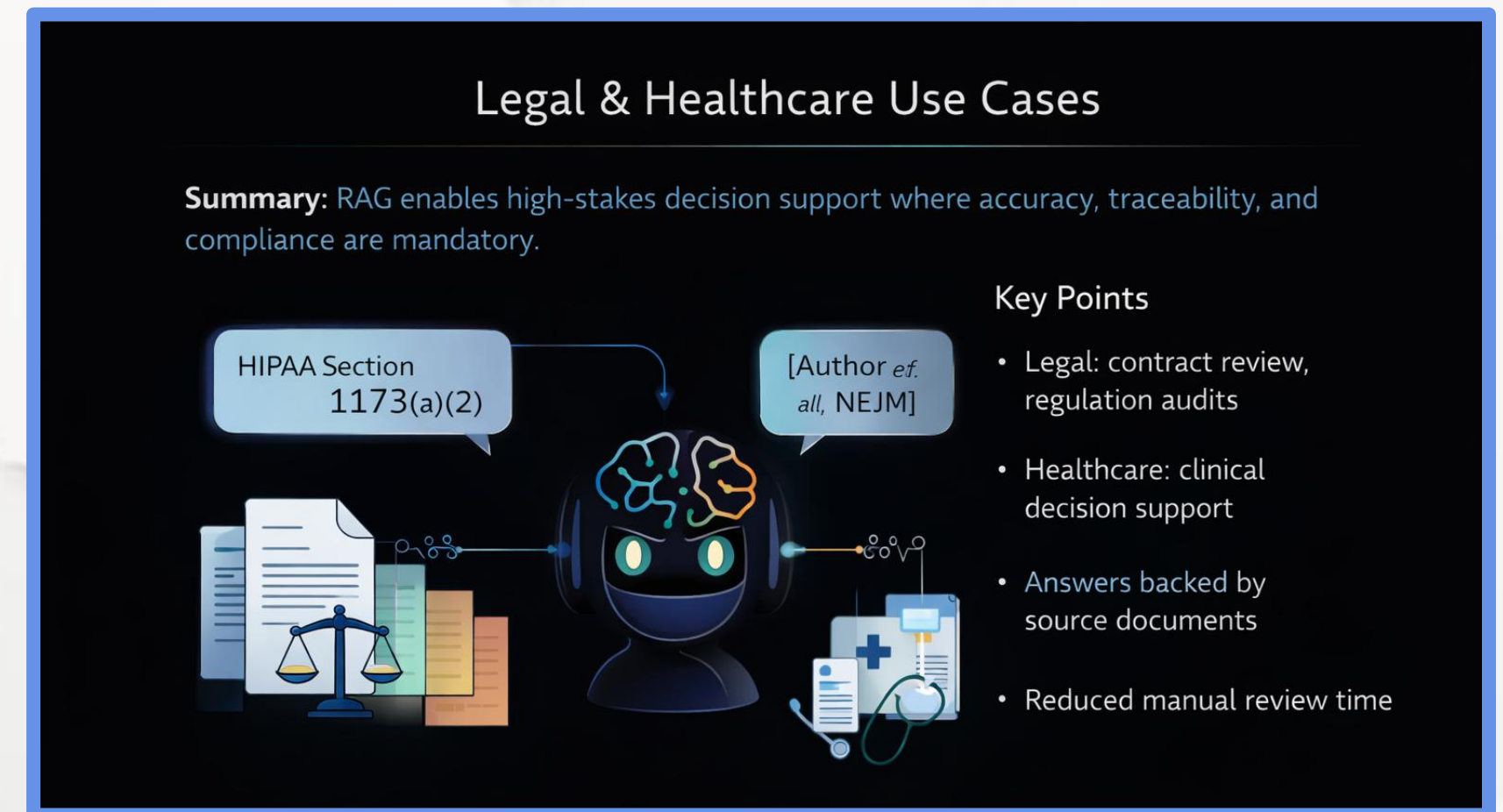


Chapter 10.3 Real-World Applications of LLMs and RAG

Legal & Healthcare Use Cases

RAG enables high-stakes decision support where accuracy, traceability, and compliance are mandatory.

- Legal: contract review, regulation audits.
- Healthcare: clinical decision support.
- Answers backed by source documents.
- Reduced manual review time.

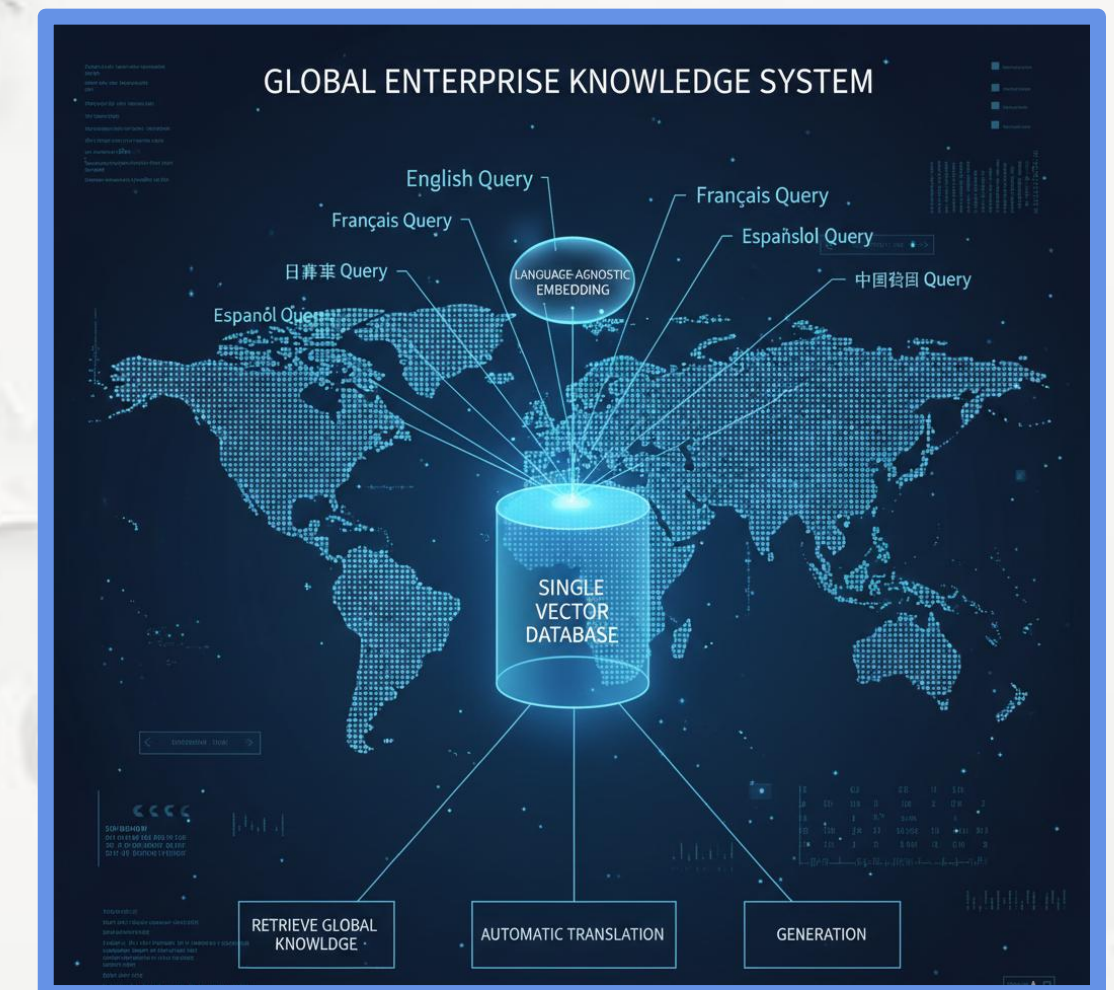


Chapter 10.3 Real-World Applications of LLMs and RAG

Multilingual Enterprise Knowledge

Modern embeddings allow organizations to access global knowledge regardless of language barriers.

- Language-agnostic embeddings.
- Query in one language, retrieve in another.
- Centralized global knowledge.
- Automatic translation at generation time.



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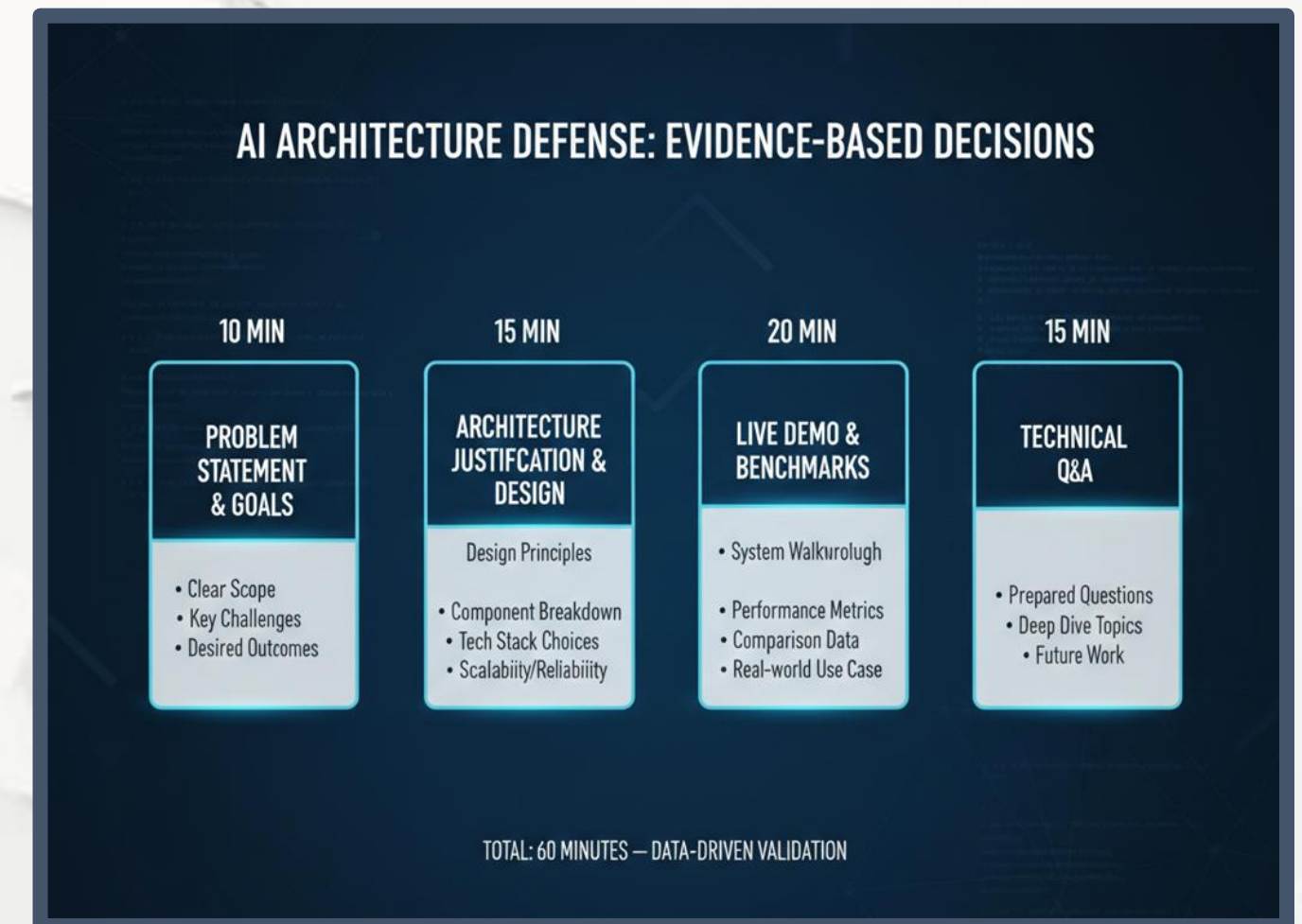


Chapter 10.4 Presentation Guidelines and Final Review

How to Present a RAG System

A professional AI presentation defends architectural decisions with evidence, not narratives.

- Clear problem statement.
- Architecture justification.
- Live demo or benchmarks.
- Prepared technical Q&A.

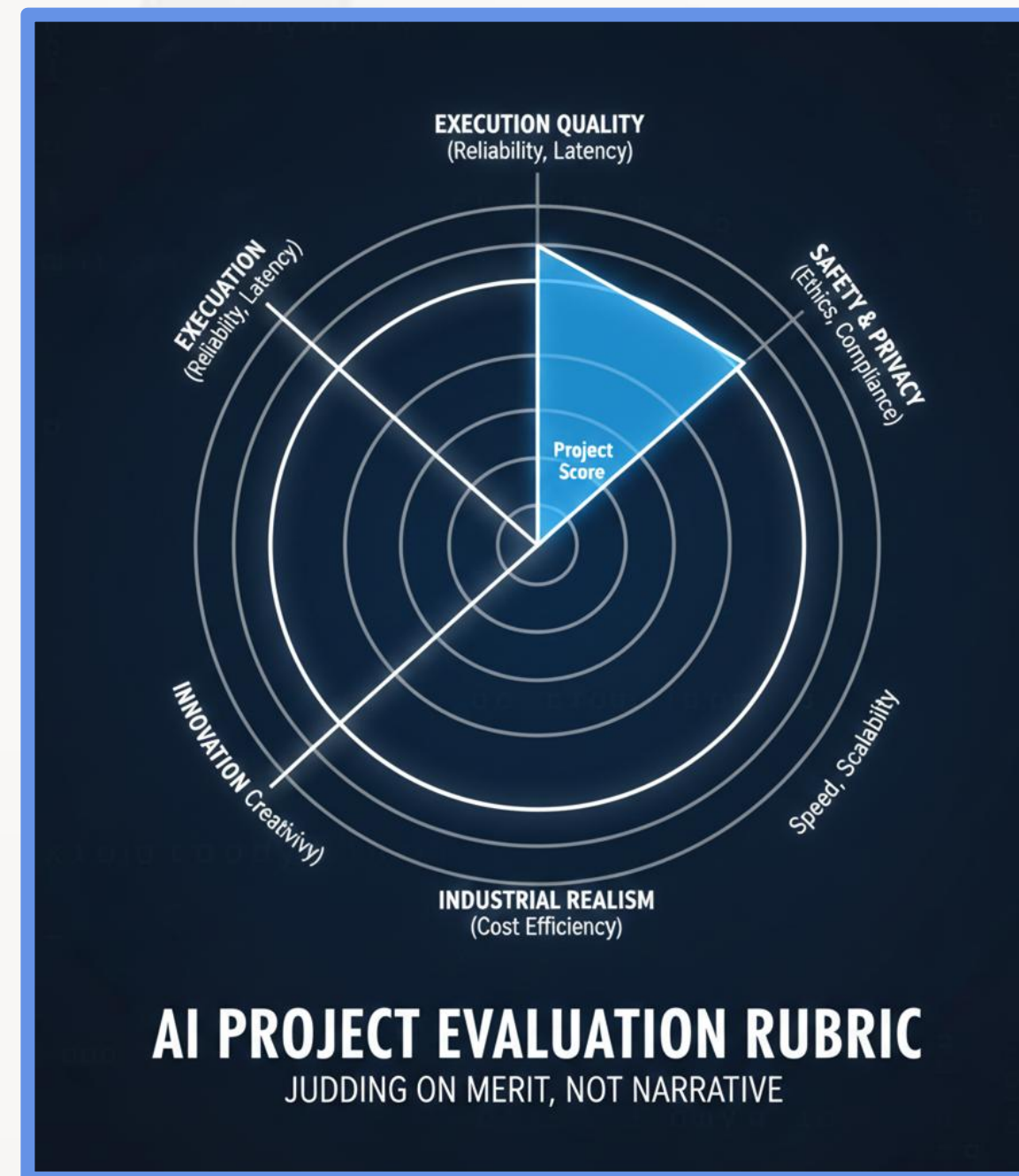


Chapter 10.4 Presentation Guidelines and Final Review

Evaluation Criteria

Projects are judged on execution quality, innovation, and industrial realism.

- Retrieval quality.
- Latency and scalability.
- Safety and privacy.
- Cost efficiency.

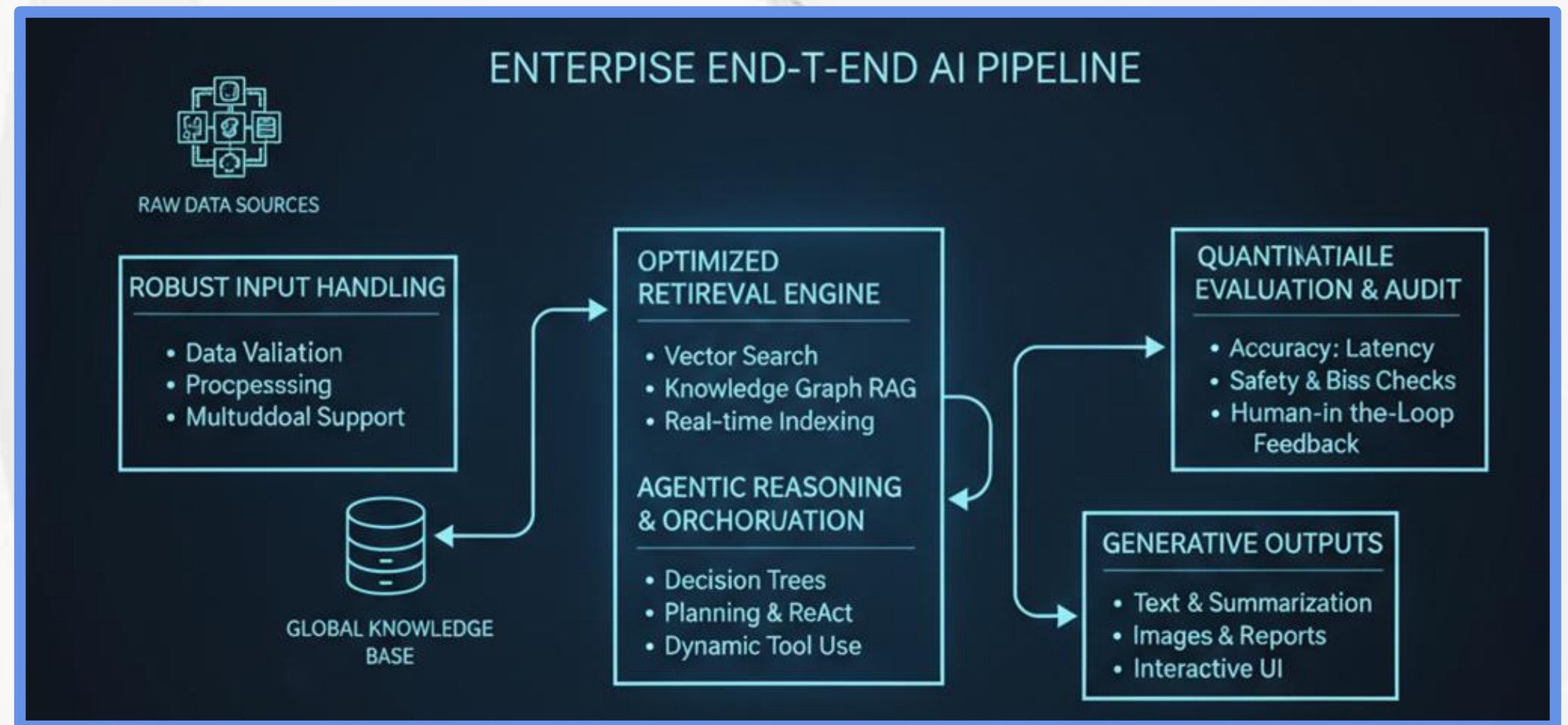


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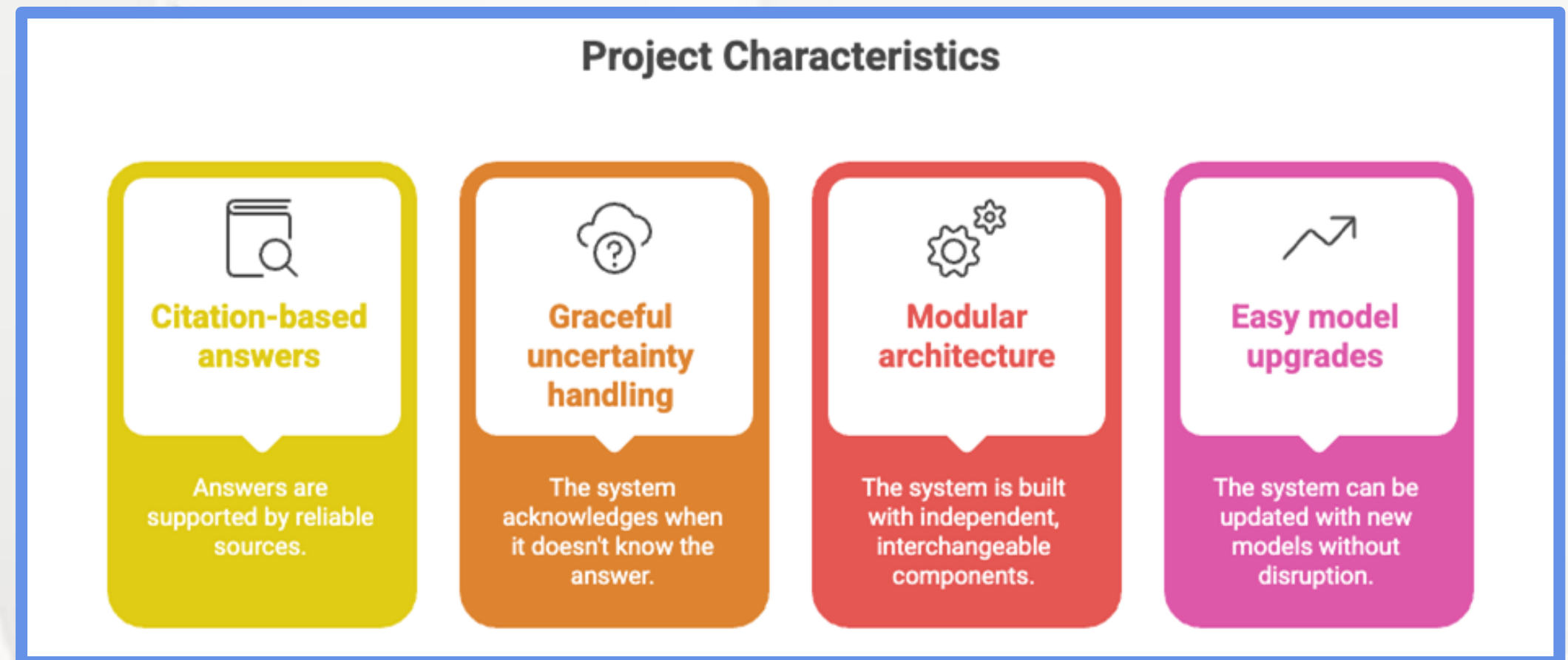


Chapter 10.5 Student Capstone Project Presentations

What Excellence Looks Like

Capstone projects demonstrate full-stack AI system thinking, not isolated techniques.

- Robust input handling.
- Optimized retrieval.
- Agentic reasoning.
- Quantitative evaluation.

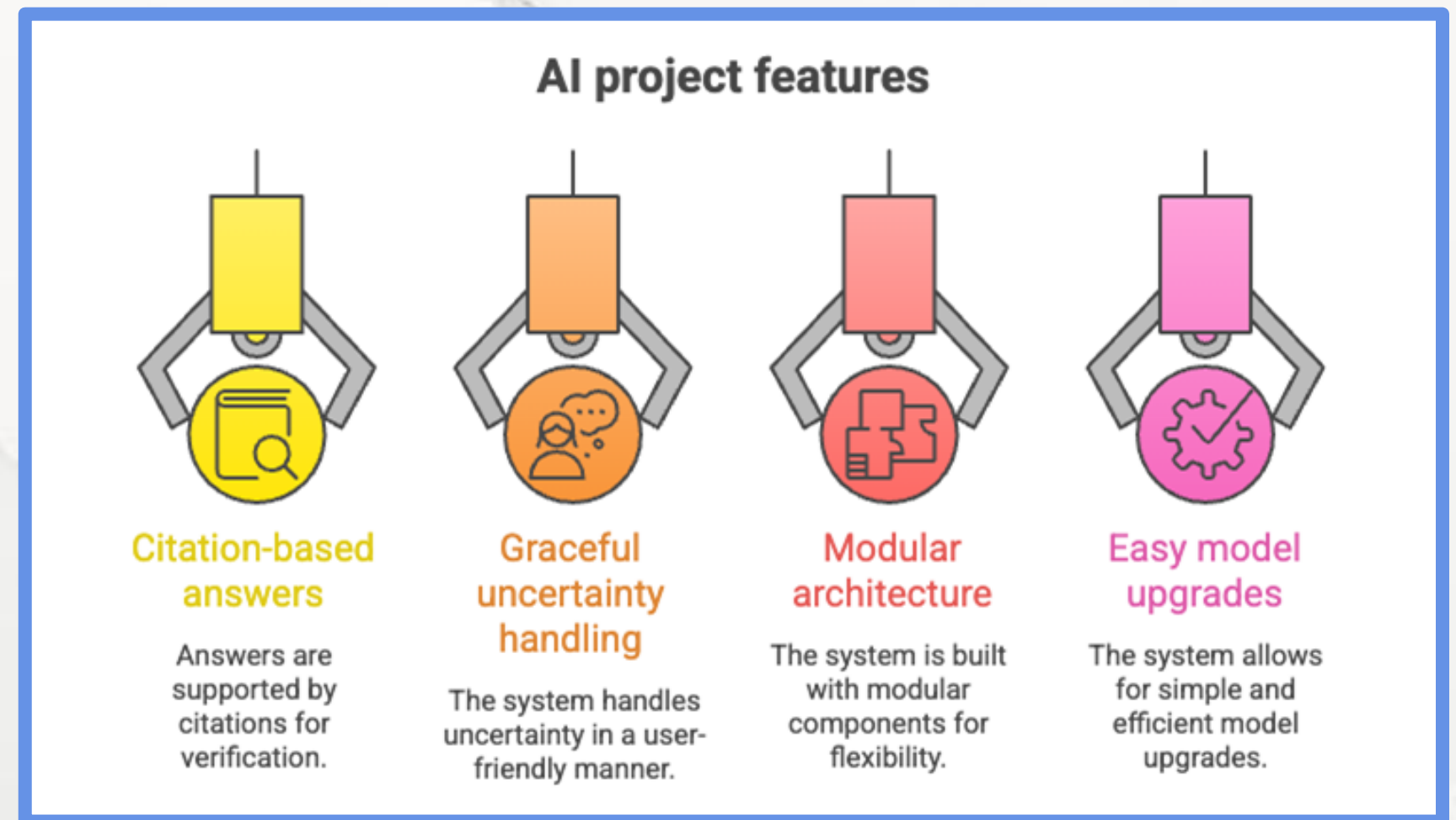


Chapter 10.5 Student Capstone Project Presentations

Production-Ready Criteria

Top projects balance factuality, usability, and maintainability.

- Citation-based answers.
- Graceful uncertainty handling.
- Modular architecture.
- Easy model upgrades.



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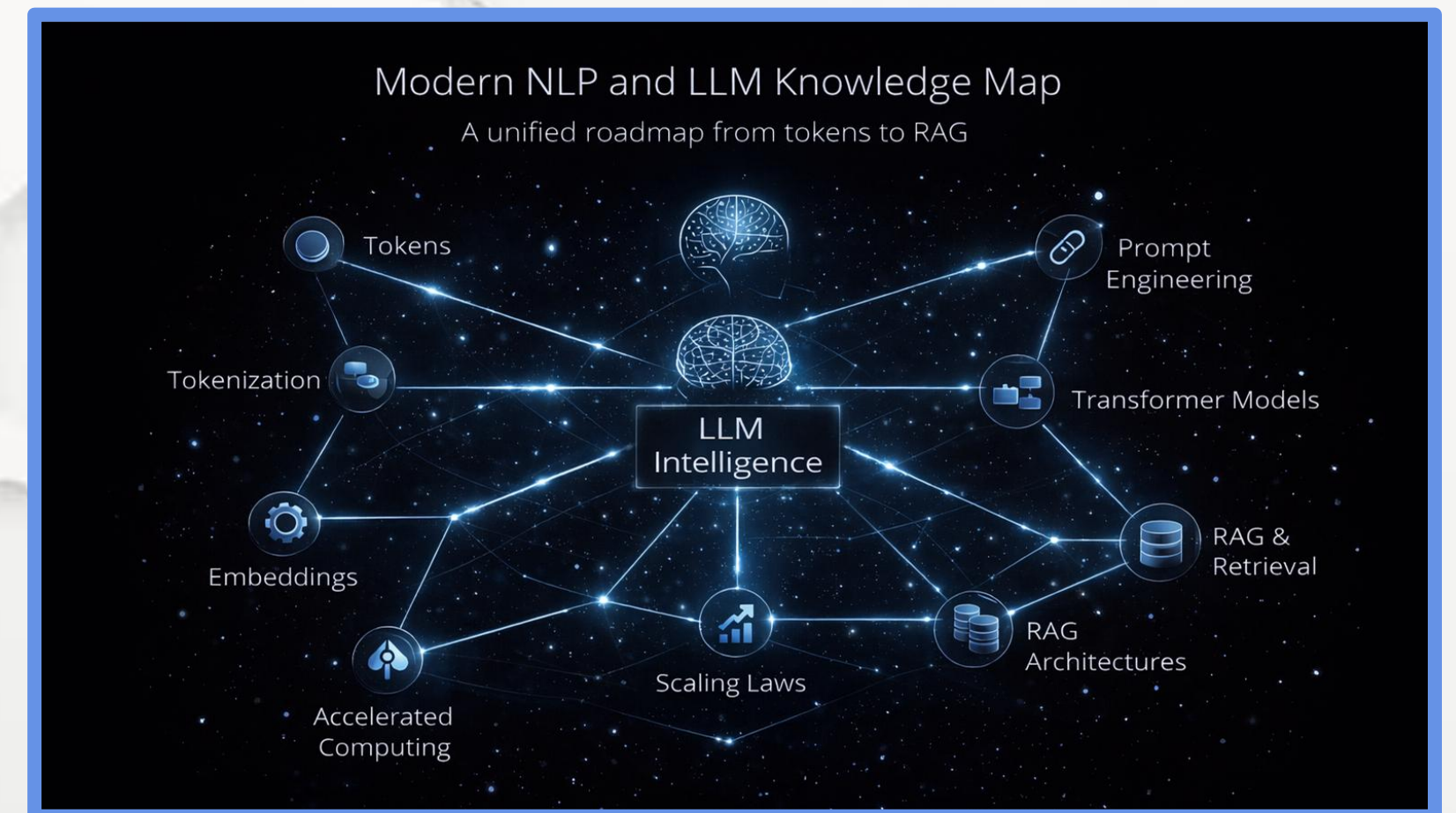


Chapter 10.6 Course Wrap-up and Next Steps

What You Have Mastered

This course covered the complete modern NLP and LLM stack, from tokens to enterprise-grade systems.

- Tokenization & embeddings
- Transformers & scaling laws
- Prompt engineering
- PEFT & LoRA
- RAG architectures



Chapter 10.6 Course Wrap-up and Next Steps

The Road Ahead

The future of AI lies in grounded, autonomous, and agentic systems that act, not just respond.

- Agentic workflows
- Multimodal reasoning
- Autonomous task execution
- Continuous learning mindset

The future of AI lies in grounded, autonomous, and agentic systems that act, not just respond.

Key Points

- Agentic workflows
- Multimodal reasoning
- Autonomous task execution
- Continuous learning mindset



References & Further Reading

- Lewis et al. (2020) — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
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- Bender et al. (2021) — On the Dangers of Stochastic Parrots.





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